

<https://gemini.google.com/share/7ccfff76d2f3>

1. Overall quality

Coverage

- It hits almost all the big themes from your discussion notes:
 - SGD vs signSGD and implicit bias, norms & initialization.
dis03_solution
 - Newton–Schulz / spectral dynamics.
dis04_solution
 - CNNs as Toeplitz matrices and parameter sharing.
dis05_solution
 - Residual blocks & gradient flow; dropout $\Leftrightarrow$ ridge.
dis06_solution
 - GNN message passing & virtual node.
dis07_solution
 - SSMs as convolutions vs recurrences and using FFT.
 - RoPE / positional encoding, argmax attention, MQA.
dis10_solution
 - Inference-time compute (beam, sampling) and soft-prompting.
dis11_solution
 - Catastrophic forgetting & LwF.
dis12_solution

Conceptually that's a **solid match** to the actual discussions.

Exam design

- Both mock exams are on-topic and look like something you *could* see in 182:
 - Questions about SGD vs signSGD geometry and implicit bias (matches Dis3 Q3).
dis03_solution
 - Convolution-as-matrix questions and parameter counts (Dis5).
dis05_solution
 - SSM kernel derivations and complexity (Dis8/9).
 - RoPE inner-products and relative positions (Dis10).
dis10_solution
 - Beam search bookkeeping and KL/LwF reasoning (Dis11/12).

Difficulty is around “midterm / final short-answer” level: not trivial, but doable if you understand the notes.

So as a **conceptual review + practice source** I'd say it's roughly **B+ / A-** quality.

2. Important mistakes you should correct

These are the ones I would *not* trust as written.

(a) Initialization scaling from RMS norm (Dis3)

Your notes show that for $x \sim N(0, I)$ and W with i.i.d. $N(0, 1)$,
 $E\|Wx\|_2^2_{\text{RMS}} = d_{\text{in}}$, and to keep that around 1 you scale W by $1/\sqrt{d_{\text{in}}}$, not $1/d_{\text{in}}$.

dis03_solution

Gemini text says “weights must be scaled by $1/d_{\text{in}}$ ”, which is wrong by a square-root. That would make the activations too small.

Fix in your head:

Proper Xavier-style scaling here is $W \sim N(0, 1/d_{\text{in}})$, i.e. multiply the raw $N(0, 1)$ matrix by $1/\sqrt{d_{\text{in}}}$.

(b) Dropout $\Leftrightarrow$ ridge regularization scaling (Dis6)

From the official derivation:

- Expected dropout objective becomes
$$\|y - pXw\|_2^2 + p(1-p)\|w\|_2^2 \approx \|y - pXw\|_2^2 + p(1-p)\|w\|_2^2.$$
- After rescaling to a standard ridge form, the effective penalty is proportional to $(1-p)/p$.

dis06_solution

Gemini says something like “penalty scales with $p/(1-p)$ ” and then also claims “as $p \rightarrow 1$, regularization $\rightarrow 0$; as $p \rightarrow 0$, $\rightarrow \infty$.” Those two statements don’t match each other or the notes.

Correct behavior:

- Small p (heavy dropout)** $\rightarrow$ penalty **large** $\propto (1-p)/p \rightarrow$ strong regularization.
- p close to 1 (little dropout)** $\rightarrow$ penalty **tiny** $\rightarrow$ almost no regularization.

So conceptually they understood “more dropout = more regularization,” but the actual formula in text is flipped.

(c) Newton–Schulz convergence range (Dis4)

The notes work this out carefully:

- $p(\sigma) = 1.5\sigma - 0.5\sigma^3$.
 - Fixed points at $\sigma = 0, \pm 1$.
 - From the cobweb diagram and derivative, singular values:
 - $0 < \sigma < \sqrt{3}$ converge to +1,
 - $\sigma = \sqrt{3}$ maps to 0,
 - $\sigma > \sqrt{3}$ diverges in magnitude (oscillating in sign).
- dis04_solution

Gemini's summary says the "basin of attraction is roughly (0,3)." That's a very loose statement and overshoots the safe region ($3 > \sqrt{5}$). As practical exam prep, you want to know the more precise story your notes give.

(d) RMS "normalization neuron" gradient (Exam 1, Q4)

For $y = \|x\|_2^2_{\text{RMS}} = (1/d) \sum x_i^2$, the correct gradient is

- $\partial y / \partial x_k = (2/d) x_k$
- $\nabla_x y = (2/d) x$

and the expected squared norm is

- $\|\nabla y\|_2^2 = (4/d^2) \sum x_k^2$
 - $E[\|\nabla y\|_2^2] = (4/d^2) \cdot d = 4/d \rightarrow 0$ as $d \rightarrow \infty$ (vanishing as $1/d$).
- dis03_solution

The solution sketch in the Gemini exam mixes up derivatives and gets inconsistent scaling (and then verbally says it "vanishes as $1/d$ " anyway). Treat that whole computation as suspect and recompute it yourself using the notes.

(e) Small imprecisions

Not fatal, but worth noting:

- **SGD vs signSGD geometry** is described correctly at a high level, but the link to "maximal update learning rate" and RMS scaling for signGD is missing; Dis4 explicitly shows you need a $1/d$ scaling on the learning rate for signGD to keep feature updates dimension-independent.
- dis04_solution

- **Newton–Schulz**: they don’t mention the specific “ $\sigma > \sqrt{5}$ diverges” point that appears in your diagram on page 3, so their stated basin is overly optimistic.
dis04_solution
- **Some exam solutions** are more hand-wavy than you’d want on a real 182 exam (e.g., not checking ranges carefully, or skipping constants in complexity).

3. Where Gemini’s write-ups *are* strong

Despite those issues, a lot is genuinely useful:

- The **conceptual arcs** (optimization $\rightarrow$ geometry; residuals as gradient highways; SSM duality; attention as solving the bottleneck; RoPE as relative position) line up very nicely with how the discussions are written.
- The mock exams:
 - Ask you to **derive kernels and Toeplitz matrices** (exactly what Dis5 emphasizes).
dis05_solution
 - Make you **unroll SSMs and write kernels** (Dis8 & 9).
 - Force you to **compute attention scores and reason about RoPE geometry** (Dis9–10).
 - Hit **beam search bookkeeping** and LwF at the level of the discussion questions.

If you actually solve those questions yourself (without trusting the solution algebra), they’re solid practice.

4. How I’d use Gemini’s material

I would:

1. **Keep both reviews** as “concept maps” of the course.
2. **Annotate them** with corrections:
 - Fix the $1/\sqrt{d_{\text{in}}}$ scaling.
dis03_solution
 - Fix dropout’s $(1-p)/p$ regularization dependence.
dis06_solution
 - Note the correct Newton–Schulz convergence ranges.
dis04_solution
 - Redo the RMS gradient question properly.
3. **Use the mock exams as practice only**, and always cross-check your derivations against the official discussion solutions afterward (the PDFs you uploaded are gold).

If you want, I can next take one of Gemini's exam questions you're unsure about and walk through the full, clean solution step-by-step so you have a "corrected key" to work from.